

Electrical Engineering B.S. Degree 2025-2026 Curriculum

Math Courses

MATH 19A

Calculus I
Prerequisites: Math Placement score of 400 or more or Math 3

MATH 19B

Calculus II
Prerequisites: Math 19A

STAT 131

Introduction to Probability Theory
Prerequisites: Math 19B.
(AM 30 or Math 23A or 23B recommended)

OR

CSE 107

Probability and Statistics for Engineers
Prerequisites: CSE 16; and AM 30 or Math 23A
NOTE: CSE 16 is not a major requirement for EE

AM 30

Multivariate Calculus for Engineers
Prerequisites: AM 10 or Math 21; Math 19B

AND

AM 100

Mathematical Methods for Engineers
Prerequisites: AM 20 or Math 24, and AM 30 or MATH 23B

OR

MATH 23A

Vector Calculus I
Prerequisites: Math 19B or 20B

AND

MATH 23B

Vector Calculus II
Prerequisites: Math 23A

AM 10*

Math Methods I
Prerequisites: Math Placement score of 400 or higher or Math 3

OR

MATH 21

Linear Algebra
Prerequisites: Math 19A

AND

AM 20*

Math Methods II
Prerequisites: Math 19B and AM 10

OR

MATH 24

Ordinary Differential Equations
Prerequisites: Math 23A

*Note: AM 10 and AM 20 are preferred.

Physics Courses

PHYS 5A/L or 15A

Mechanics
Prerequisites: Math 19A or 20A, and concurrent enrollment in Math 19B

PHYS 5B/M

Waves & Optics
Prerequisites: Phys 5A/L and Math 19B

PHYS 5C/N or 15C

Electricity & Magnetism
Prerequisites: Phys 5A/L, Math 19A, Corequisite Math 19B

PHYS 5D

Heat & Thermodynamics
Prerequisites: Phys 5A/L, Phys 5B, Math 19B

Computer Engineering Courses

CSE 20

Beginning Programming in Python

OR

CSE 30

Programming Abstractions: Python
Prerequisites: CSE 20 and MATH 3 or 19A or Math placement score of 400 or more

CSE 12

Computer Systems & Assembly Language and Lab
Prerequisites: CSE 20

ECE 13

Computer Systems & C Programming
Prerequisites: CSE 12 and CSE 20
CSE 30 is recommended

Electrical Engineering Courses

ECE 80T^

Modern Technology

^This course is waived for transfer students.

ECE 102/L

Properties of Materials
Prerequisites: Phys 5A/L, 5B/M & 5C/N

ECE 103/L

Signals and Systems
Prerequisites: ECE 101/L and AM 20 or Math 24

ECE 135/L

Electromagnetic Fields and Waves
Prerequisites: ECE 101/L, and Math 24 or AM 20 or Phys 116A

ECE 151

Communications Systems
Prerequisites: ECE 101/L, ECE 103, and STAT 131 or CSE 107

Electronics Courses

ECE 101/L

Electronic Circuits
Prerequisites: Phys 5C/N and Math 24 or Phys 116A or concurrent or previous enrollment in AM 20

CSE 100/L

Logic Design
Prerequisites: CSE 12
Requires declared major status

ECE 171/L

Analog Electronics
Prerequisites: ECE 101/L

Elective Requirements: In addition to the above, Electrical Engineering majors must complete 4 additional upper-division courses (minimum of 3 courses from one track). Unlisted graduate-level courses may be used to fulfill an elective requirement with prior department approval. **Most elective courses have additional prerequisites. They are subject to change frequently. Please visit <https://catalog.ucsc.edu/current/General-Catalog/Courses/ECE-Electrical-and-Computer-Engineering> to ensure you have met them.**

Design Elective: One of the four concentration courses chosen must include at least one of the following design electives: ECE 118, ECE 121, ECE 157/L, ECE 167, or ECE 173. *The design elective must be taken before ECE 129A.*

Design Elective:

Concentration Elective #1:

Concentration Elective #2:

Concentration Elective #3:

Comprehensive Requirement (ECE129ABC or ECE 129A & ECE 195):

ECE 129A Capstone Project I

Prerequisites: ECE171/L, CSE 100/L, and at least one design elective.

ECE 129B Capstone Project II

• Prerequisites: ECE 129A

ECE 129C Capstone Project III

• Prerequisites: ECE 129B

ECE 195* (10 units)

Senior Thesis

Prerequisites: ECE 129A

*Requires department approval. Ask BE Advising for more information

Exit Requirements:

1. Exit Survey
2. Exit Interview
3. Maintain a 2.5 cumulative GPA in all required and elective courses for the major, OR submit a portfolio for department review, OR submit a senior thesis with department approval.

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Fall _____	Winter _____	Spring _____	Summer _____

Fall _____	Winter _____	Spring _____	Summer _____

Fall _____	Winter _____	Spring _____	Summer _____

Fall _____	Winter _____	Spring _____	Summer _____

Note:

The Electrical Engineering Major Qualification requirements require all courses in the darker shaded boxes and at least one of the courses in the lighter shaded boxes. All courses on this chart are required for the Electrical Engineering B.S. degree. Major Qualification requirements can be found on the Prepare to Declare charts on the BE Undergraduate Advising website.

The prerequisites listed on this curriculum chart are accurate as of July 21, 2025 according to UCSC's general catalog. Prerequisites listed on this chart are subject to change and students should refer to the catalog for the most up to date requirements.

Students interested in Honors Physics (Physics 15A, 15C) should speak with BE undergraduate advising.